

Channel-Robust RF Fingerprint Identification for Multi-Antenna 5G User Equipments

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Abstract—Radio frequency fingerprint (RFF) is a promising solution for realizing secure and efficient device identification. However, the accuracy of currently existing solutions suffer from multipath effects in practical scenarios. In this paper, we provide a robust RFF identification method that leverages channel state information (CSI) feedback to counteract the effect of the channel on the extracted RFF features. A straightforward zero-forcing (ZF) equalization fails to fully decouple RF impairments from the channel, making conventional approaches ineffective. To overcome this challenge, we utilize the potential of multi-antenna and introduce a new device-specific feature called Relative-RFF (R-RFF), which represents the relation between different RF chains in a multi-antenna transmitter. We propose an enhanced ZF post-equalization algorithm to eliminate the multipath channels and preserve the users' R-RFF to the greatest extent. We evaluate the robustness of R-RFF under various channel conditions and noise levels and the performance of R-RFF in terms of identification accuracy under different channel scenarios. The results show that the proposed R-RFF method can achieve an identification accuracy of 91.2% for 70 devices in tapped delay line channel with a signal-to-noise ratio (SNR) of 30 dB.

Index Terms—Physical-layer security, radio frequency fingerprint, device identification, multipath channels.

I. INTRODUCTION

WITH the remarkable proliferation of wireless technologies and the widespread application of fifth generation mobile networks (5G) technology, the number of wireless devices has dramatically increased. Emerging applications such as IoT, smart cities, and autonomous driving have posed heightened demands for security and identity verification [1]. However, owing to the broadcast nature of radio signal propagation, wireless communications are vulnerable to malicious attacks. Traditional identification processes relying on digital cryptographic techniques [2] are insufficient to guard against spoofing and impersonation when a digital security system is compromised [3]. Additionally, the swift increase in the number of devices complicates the process of key generation, distribution, and management [4]. As a result, there is an urgent need for a lightweight device identification method.

Radio frequency fingerprint (RFF) represents a physical-layer solution for secure device identification [5]. During the manufacturing process, electronic components and printed circuit boards may present some impairments that cause specific signal distortion or perturbation. By utilizing these hardware impairments, it is possible to extract the RFF of a device for identification purposes. Before initiating data exchange, wireless communication systems can enhance security by detecting a device's RFF to verify its identity. RFF has been investigated for several wireless systems, including Wi-Fi [6], [7], [8], LoRa [9], [10], [11], ZigBee [12], [13], long-term evolution (LTE) [14], [15], [16], all of which have proved the effectiveness of RFF. However, there has been limited exploration of RFF within 5G new radio (NR) systems. RFF for 5G user equipment (UE) faces challenges from the following two aspects. First, 5G leverages complex reference signal mechanisms which make it complicated to select and schedule reference signals to realize RFF. Second, as 5G is a wideband system, the transmissions over multipath channels present a significant challenge for RFF extraction. Multipath fading can lead to signal fading and distortion, overlapping and distorting the original RFF signal. Delay spread blurs the details and features of the signal, making the temporal analysis during RFF more complex. Different paths in the multipath

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TABLE I
A COMPARISON BETWEEN EXISTING RFF SCHEMES AND THE PROPOSED R-RFF SCHEME

Scheme	Protocol	Authentication timing	Antenna configuration	Categories	Key assumptions
Yin <i>et al.</i> [16]	LTE	initial authentication	SISO	channel-robust RFF	stable channel conditions
Kandel <i>et al.</i> [20]	802.11n	initial authentication	MIMO	channel-robust RFF	high CSI quality
Soltani <i>et al.</i> [21]	802.11a	initial authentication	SISO	data augmentation	large-scale data required
Xing <i>et al.</i> [25]	802.11n	initial authentication	SISO	decoupling RFF	RFF feature loss allowed
Proposed R-RFF	5G NR	initial & periodic authentication	MIMO	channel-robust RFF, comprehensive simulation & decoupling RFF	multiple antennas required

channels can introduce phase differences, leading to phase offsets in the received signal. All of these severely reduce the accuracy and reliability of the fingerprint [6], [7]. Therefore, designing a channel-robust RFF scheme is highly desirable.

According to different principles of realizing channel-robust RFF, the current research efforts can be divided into three categories.

- *Channel-Robust RFF Extraction Methods:* Methods include RF-DNA [17], signal-to-noise-ratio (SNR) trace in the sector-level sweep (SLS) procedure [18], channel independent spectrogram [19], phase relation [20], and DCTF [16], among others.
- *Data Augmentation Methods:* Strategies involve exposing a classifier to a broad range of channels [21], leveraging the original samples as supervisory information for augmentation [22], DeepCRF [23], and slicing-enhanced preprocessing incorporating noise augmentation [24].
- *Decoupling RFF Methods:* Methods include different spectrums of adjacent signal symbols [25], nonlinearity estimation [26], ESPRIT [27], DeepRadioID [28], STBC reconstruct channel [29], and FID [30].

To sum up, Table I presents representative research across the aforementioned three categories. It compares several existing RFF schemes in multipath channels with our proposed relative-RFF (R-RFF) scheme. In terms of authentication timing, prior works primarily focus on initial authentication, whereas the proposed R-RFF scheme supports both initial and periodic authentication, enabling enhanced security over time. Regarding antenna configuration, existing research primarily focused on single antenna RFF methods resilient to multipath interference, neglecting the novel dimensions brought by the growing deployment of multiple antennas. From the key assumption perspective, single-antenna channel-robust RFF methods often assume stable channel conditions or high CSI quality, which may not hold in dynamic environments, leading to limited recognition performance. Data augmentation approaches rely heavily on large datasets, while decoupling-based RFF methods introduce RFF feature loss. Multiple antennas systems offer the potential to separate the RFF from the channel. Existing multi-antenna RFF studies primarily focus on either signal subspace reconstruction at the receiver [27] or feature extraction from different antenna chains at the transmitter [31]. However, the method in [27] requires at least $2N + 1$ receiving antennas for effective RF impairment separation, limiting its practicality.

In this paper, we address the challenging issues of multipath interference in RFF within 5G systems by leveraging multi-

antenna solutions. To combat multipath effects, this paper proposes channel-robust RFF features and introduces comprehensive simulation to enhance security in 5G systems. Our main contributions are summarized as follows.

- We introduce a new RFF feature for multi-antenna devices, named relative-RFF, designed for channel-robust identification. This feature is derived from the unique and stable relationship of RF impairments across the different antenna chains within a single device, providing a new dimension for physical layer security.
- We design a framework to extract channel-robust R-RFF, designing an enhanced zero-forcing (ZF) post-equalization function for the framework, and propose the process for constructing the function. Furthermore, we provide a theoretical analysis to demonstrate that the extracted R-RFF is independent of the channel state, ensuring its robustness against multipath effects.
- We establish a 5G RFF simulation system for performance evaluation. We simulate six types of channels with a different number of paths in MATLAB using 5G new radio (5G NR) physical-layer (PHY) frames for 110 UEs and different SNR levels. Simulation results show that the proposed method can achieve 91.2% classification accuracy for 70 devices in the tapped delay line (TDL) channel of the 3rd generation partnership project (3GPP) technical report (TR) 38.901 [32] with SNR of 30 dB.

In this paper, we expand on our previous work from [33]. First, we introduce a framework for extracting channel-robust RFF and design a post-equalization function. We provide a theoretical analysis demonstrating that R-RFF is independent of the channel state, ensuring its robustness against multipath effects. Furthermore, we conduct more comprehensive simulations, covering all TDL model scenarios to evaluate RFF performance under diverse multipath conditions and enhance generalization capability.

The rest of the paper is organized as follows. Section II describes the system model and the RFF system. Section III elaborates on the multipath challenges and proposes the framework for extracting channel-robust RFF. Section IV illustrates the multi-antenna channel-robust RFF extraction methods. Section V outlines the training and identification process. Section VI presents the simulation results. Section VII discusses the limitations of this study and future work. Finally, Section VIII concludes this paper.

II. SYSTEM MODEL

In this paper, we consider an orthogonal frequency division multiplexing (OFDM) time division duplexing (TDD) system

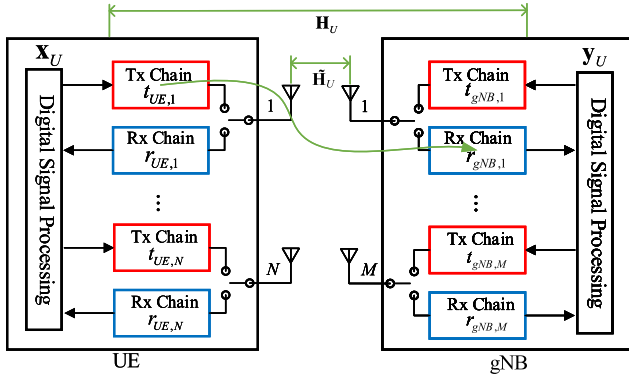


Fig. 1. The uplink and downlink channels between UE and gNB.

with multi-antenna UEs in a 5G system. We consider a next Generation Node B (gNB) with M antennas and K UEs, each equipped with N antennas ($M > N$). The legitimate UE sends a pilot signal to the gNB for device identification. Similarly, the rogue device has N' antennas and sends a pilot signal in an attempt to connect to the gNB. All signals transmitted from UEs to the gNB propagate through multipath channels. The system operates in TDD mode, assuming reciprocity between the uplink and downlink channels.

A. RF Impairments and Channel Model

These impairments arise from a combination of manufacturing deviations in RF components. To facilitate the separation between RF impairments and channel characteristics, we model the uplink and downlink channels from a frequency domain perspective. We assume that the received channel estimation is influenced by three primary factors: (i) the frequency response of the transmit hardware, (ii) the real physical wireless channel, and (iii) the frequency response of the receive hardware. The relation from transmitter to receiver can be mathematically represented as [34]:

$$\mathbf{H} = \mathbf{R}\tilde{\mathbf{H}}\mathbf{T}, \quad (1)$$

where $\mathbf{H}$ denotes the effective channel matrix, $\tilde{\mathbf{H}}$ represents the true channel matrix, $\mathbf{R}$ corresponds to the diagonal matrix of the receive hardware impairments, and $\mathbf{T}$ corresponds to the diagonal matrix of the transmit hardware impairments. The impact of these hardware impairments manifests in several ways. Key among them is oscillator imperfections, which include carrier frequency offset (CFO) and phase noise, as well as in-phase and quadrature (IQ) imbalance. Additionally, the nonlinearity of power amplifiers also plays a significant role.

Let $\mathbf{H}_U \in \mathbb{C}^{M \times N}$ and $\mathbf{H}_D \in \mathbb{C}^{N \times M}$ be the uplink and downlink channels between gNB and UE, respectively. As shown in Fig. 1, which depicts the individual transmit (red blocks) and receive (blue blocks) chains, the equivalent channel from the UE to the gNB can be modelled as

$$\mathbf{H}_U = \mathbf{R}_{gNB}\tilde{\mathbf{H}}_U\mathbf{T}_{UE}, \quad (2)$$

where $\tilde{\mathbf{H}}_U$ is the real uplink wireless channel between UE and gNB, $\mathbf{R}_{gNB} = \text{diag}(r_{gNB,1}, r_{gNB,2}, \dots, r_{gNB,M})$ is the receiving chain RF impairment of gNB, and $\mathbf{T}_{UE} =$

 TABLE II
SUMMARY OF KEY NOTATIONS

Symbol	Definition
N, M	Number of antennas at the UE and gNB.
$\tilde{\mathbf{H}}_U, \tilde{\mathbf{H}}_D$	True uplink and downlink channel matrices.
$\hat{\mathbf{H}}_U, \hat{\mathbf{H}}_D$	Estimated uplink and downlink channel matrices.
$\mathbf{T}_{UE}, \mathbf{R}_{UE}$	UE's transmit and receive RF impairment matrices.
$\mathbf{T}_{gNB}, \mathbf{R}_{gNB}$	gNB's transmit and receive RF impairment matrices.
$\mathbf{C}_{UE}$	UE's intrinsic R-RFF matrix.
$\mathbf{C}_{gNB}$	gNB's intrinsic R-RFF matrix.
$\mathbf{C}_{gNB}^r$	gNB's measurable relative impairment matrix.
$\mathbf{W}$	Pseudo-inverse matrix.
F	The post-equalization function.

$\text{diag}(t_{UE,1}, t_{UE,2}, \dots, t_{UE,N})$ is the transmitting chain RF impairment of UE. Similarly, the equivalent downlink channel can be modelled as

$$\mathbf{H}_D = \mathbf{R}_{UE}\tilde{\mathbf{H}}_D\mathbf{T}_{gNB}, \quad (3)$$

where $\tilde{\mathbf{H}}_D$ is the real downlink wireless channel between UE and gNB, $\mathbf{R}_{UE} = \text{diag}(r_{UE,1}, r_{UE,2}, \dots, r_{UE,N})$ is the receiving chain RF impairment of UE, and $\mathbf{T}_{gNB} = \text{diag}(t_{gNB,1}, t_{gNB,2}, \dots, t_{gNB,M})$ is the transmitting chain RF impairment of gNB. Due to the reciprocity of uplink and downlink channels within coherence time, the wireless channel $\tilde{\mathbf{H}}_U = \tilde{\mathbf{H}}_D^T$.

B. RFF System Overview

The gNB's objective is to identify the signal-transmitting device using extracted RFF features from the received signal. A generic RFF framework primarily comprises two stages: training and classification.

- During the training stage, the receiver, acting as an authenticator, collects packets from each training device. The receiver then applies signal preprocessing algorithms to these collected packets, including packet detection and synchronization, CFO estimation and compensation. The critical step involves extracting the RFF features from the processed data for device classification. These RFF features extracted from the data packets are then used to train a model for device classification. The training process is a one-time, offline operation conducted in a secure, attacker-free environment.
- In the classification stage, the devices to be identified emit wireless signals, and the receiver conducts signal preprocessing and RFF extraction. Finally, the identity of the signal source is classified based on the extracted features and the outputs of the pre-trained classifier model.

According to Section II-A, the hardware impairments of UE, multipath channels, and gNB hardware impairments are coupled together. It is necessary to design a channel-robust feature extraction method for UEs to achieve decoupling from multipath channels and accurately identify UEs. For clarity and easy reference, the key notations used throughout this paper are summarized in Table II.

III. ROBUST MULTI-ANTENNA R-RFF FEATURES

In this section, we first state the challenges of obtaining channel-robust features in multipath channels. Subsequently,

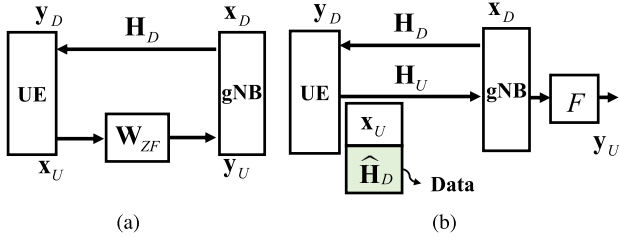


Fig. 2. (a) Intuitive idea of RFF extraction based on precoding. (b) RFF extraction framework based on post-equalization.

we present our framework that integrates post-equalization and interaction protocols to extract channel-robust RFF.

A. The Challenges of Obtaining Channel-Robust Features

The intuitive idea to mitigate the impact of channel effects on RFF is to use channel reciprocity and precoding to eliminate multipath channels in RFF. This method is also widely used in communication systems to eliminate the effect of the channel on data transmission in multiple-input multiple-output (MIMO) systems [35].

The precoding scheme is shown in Fig. 2(a). Assume that gNB sends pilot signal $x_D \in \mathbb{C}^{M \times 1}$ to UE, and UE receives

$$y_D = H_D x_D + z_D, \quad (4)$$

where, $H_D \in \mathbb{C}^{N \times M}$ represents the downlink equivalent channel, $x_D \in \mathbb{C}^{M \times 1}$ represents the vector of the downlink pilot signals from gNB, and $z_D = [z_1, z_2, \dots, z_N]^T$ represents the additive white Gaussian noise (AWGN), $z_n \sim \mathcal{N}(0, \sigma_n^2)$. The signal with precoding after the receiving chain from UE to gNB is given by

$$y_U = H_U W_{ZF} x_U + z_U, \quad (5)$$

where, $H_U \in \mathbb{C}^{M \times N}$ represent the uplink equivalent channel, W_{ZF} represents the ZF precoding, $x_U \in \mathbb{C}^{N \times 1}$ represents the vector of the uplink pilot signals from UE, and $z_U = [z_1, z_2, \dots, z_M]^T$ represents the AWGN, $z_m \sim \mathcal{N}(0, \sigma_m^2)$.

The precoding matrix W_{ZF} is given by

$$\begin{aligned} W_{ZF} &= (\hat{H}_D^T)^H \left[\hat{H}_D^T (\hat{H}_D^T)^H \right]^{-1} \\ &= \hat{H}_D^* \left(\hat{H}_D^T \hat{H}_D^* \right)^{-1}, \end{aligned} \quad (6)$$

where, $\hat{H}_D$ denotes the result of downlink channel estimation, defined as

$$\hat{H}_D = y_D (x_D)^\dagger, \quad (7)$$

and $(\cdot)^\dagger$ represents the pseudoinverse matrix. Similarly, $\hat{H}_U$ denotes the result of uplink channel estimation, defined as

$$\hat{H}_U = y_U (x_U)^\dagger. \quad (8)$$

We expand (5) as

$$\begin{aligned} y_U &= H_U \hat{H}_D^* \left(\hat{H}_D^T \hat{H}_D^* \right)^{-1} \tilde{x}_U + z_U \\ &= R_{gNB} \tilde{H}_U T_{UE} \left[T_{gNB}^T \tilde{H}_D^T R_{UE}^T \right]^{-1} \tilde{x}_U + z_U. \end{aligned} \quad (9)$$

As can be seen from (9), the precoding matrix can only eliminate the channel effects when $R_{gNB} = T_{gNB}^T$ and $T_{UE} = R_{UE}^T$. In other cases, due to the non-reciprocity of the uplink

and downlink channels, $R_{gNB} \tilde{H}_U T_{UE} \neq T_{gNB}^T \tilde{H}_D^T R_{UE}^T$, the channel effects cannot be mitigated. In addition, the RF impairments of the transmitter and the receiver are still coupled with the channel, which is a great challenge to obtain the RFF.

These precoding limitations necessitate a different strategy, leading to our proposed post-equalization framework for decoupling RFF from multipath channel.

B. The Framework for Extracting Channel-Robust RFF

In uplink transmission, the signal passes through the UE's transmitter impairments matrix T_{UE} , then encounters the channel effects $\tilde{H}_U$, and subsequently passes through the gNB's impairments matrix R_{gNB} . Since the precoding scheme uses right-multiplication rather than left-multiplication, it cannot mitigate the channel effects. Therefore, we adopt a left-multiplication post-equalization scheme.

The new framework of the post-equalization scheme is shown in Fig. 2(b). The gNB and UE estimate the channel using the reference signals sent during the coherence time to obtain $\hat{H}_D$ and $\hat{H}_U$. The UE sends $\hat{H}_D$ to gNB as data, and the gNB employs the post-equalization function F to extract channel-robust RFF.

Assume that the receiving signal after post-equalization at the gNB side is given by

$$y_U = F(\tilde{H}_U) R_{gNB} \tilde{H}_U T_{UE} x_U + z_U, \quad (10)$$

where $F(\tilde{H}_U)$ is a post-equalization function of $\tilde{H}_U$ and it is designed to eliminate the impact of multipath channels. Our objective is to identify a function F such that $F R_{gNB} \tilde{H}_U$ becomes independent of the channel. The inverse matrix of $\tilde{H}_U$ is required in F to eliminate $\tilde{H}_U$. However, it is unable to directly obtain $\tilde{H}_U$ at the gNB. The reciprocity of the uplink and downlink channels ensures that $\tilde{H}_D^T = \tilde{H}_U$. Therefore, the downlink channel estimation $\hat{H}_D$ needs to be transmitted as part of the data to the gNB to form part of $F(\tilde{H}_U)$.

The design of F is not straightforward. Given that we aim to construct a transformation that mitigates the effects of multipath channels and hardware impairments, we define F as:

$$F(\tilde{H}_U) = (T_{gNB}^T \tilde{H}_D^T R_{UE}^T)^{-1} = (T_{gNB}^T \tilde{H}_U R_{UE}^T)^{-1}. \quad (11)$$

This function attempts to invert the impact of the transmit and receive hardware impairments as well as the multipath channel. Applying F to the observed uplink channel $\tilde{H}_U$, we obtain:

$$F(\tilde{H}_U) R_{gNB} \tilde{H}_U = (T_{gNB}^T \tilde{H}_U R_{UE}^T)^{-1} R_{gNB} \tilde{H}_U. \quad (12)$$

However, the inverse matrix of $\hat{H}_D^T$ introduces the $(T_{gNB}^T)^{-1}$. Since $(T_{gNB}^T)^{-1}$ and R_{gNB} are not directly cancellable, they remain situated between $(\tilde{H}_D^T)^{-1}$ and $\tilde{H}_U$. This structural misalignment prevents the complete elimination of multipath effects, thereby making it impossible to fully decouple the device-dependent features from the channel distortions.

IV. MULTI-ANTENNA CHANNEL-ROBUST RFF FEATURES EXTRACTION METHOD

In this section, we propose a channel-robust RFF features extraction method. We first introduce the design of F and establish a connection between $(\mathbf{T}_{gNB}^T)^{-1}$ and $\mathbf{R}_{gNB}$ by introducing the new concept of R-RFF. Following this, we present an overall process for extracting channel-robust RFF features from multi-antenna devices.

A. The Design of Post-Equalization F

Our objective is to find a function F such that $F\mathbf{R}_{gNB}\tilde{\mathbf{H}}_U$ becomes independent of the channel. In Section III-B, we identify two key challenges in mitigating the impact of multipath channels on RFF. (1) It is necessary to design a factor that can transform and cancel out $(\mathbf{T}_{gNB}^T)^{-1}$ and $\mathbf{R}_{gNB}$. (2) We need to utilize CSI feedback to eliminate multipath channels and extract stable UE RFF features.

lemma 1: When the post-equalization function $F = \mathbf{W}\mathbf{C}_{gNB}$, where $\mathbf{W} = (\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T)^{-1} \hat{\mathbf{H}}_D^*$, and $\mathbf{C}_{gNB} = \mathbf{T}_{gNB}(\mathbf{R}_{gNB})^{-1}$, it can eliminate the impact of channel if the noise approaches zero approximately.

Proof: [Proof of lemma]

$$\begin{aligned} F\hat{\mathbf{H}}_U &\approx \mathbf{W}\mathbf{C}_{gNB}\hat{\mathbf{H}}_U \\ &= (\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T)^{-1} \hat{\mathbf{H}}_D^* \mathbf{C}_{gNB} \hat{\mathbf{H}}_U \\ &= (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{R}_{UE} \tilde{\mathbf{H}}_D \mathbf{T}_{gNB})^T)^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} (\mathbf{R}_{gNB})^{-1}) (\mathbf{R}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}) \\ &= \mathbf{T}_{UE} (\mathbf{R}_{UE})^{-1}. \end{aligned} \quad (13)$$

As $\mathbf{T}_{UE}$ and $(\mathbf{R}_{UE})^{-1}$ respectively represent the transmission and reception RF impairments of UE, which are independent of channels, this completes the proof of the lemma. The detailed mathematical derivation demonstrating how $F\hat{\mathbf{H}}_U$ approximates $\mathbf{T}_{UE}(\mathbf{R}_{UE})^{-1}$ is provided in Appendix A. ■

According to (13), the features obtained after post-equalization function F are channel-robust, but only contain the RF impairment of UE. We define it as the R-RFF of UE

$$\mathbf{C}_{UE} = \mathbf{T}_{UE}(\mathbf{R}_{UE})^{-1}. \quad (14)$$

However, the function F itself requires knowledge of the gNB's own R-RFF $\mathbf{C}_{gNB}$. Therefore, before we can extract the UE's fingerprint, we must first determine an achievable form of $\mathbf{C}_{gNB}$. This process is detailed in the next section.

B. Process for Finding an Achievable $\mathbf{C}_{gNB}$

The designed post-equalization function F is proved to be successful in eliminating the impact of multipath channels and preserve the R-RFF of UE in Section IV-A. In this section, we introduce the process for finding an achievable $\mathbf{C}_{gNB}$ to constitute F and extract R-RFF features, including obtaining $\mathbf{C}_{gNB}$, signal preprocessing, and obtaining pseudo-inverse matrix $\mathbf{W}$.

Algorithm 1 The Process of Obtaining R-RFF of gNB

Require: The pilot signal $\mathbf{s}_{gNB,1}, \dots, \mathbf{s}_{gNB,M}$

Ensure: The R-RFF of gNB, $\mathbf{C}'_{gNB}$

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1: for each  $m \in [2, M]$  do
2:   1-st antenna sends pilot signal  $\mathbf{s}_{gNB,1}$ ;
3:    $m$ -th antenna receives  $\mathbf{y}_{gNB,1 \rightarrow m}$ ;
4:   gNB estimates channel  $\hat{h}_{gNB,1 \rightarrow m} = \mathbf{y}_{gNB,1 \rightarrow m} (\mathbf{s}_{gNB,1})^\dagger$ ;
5:    $m$ -th antenna sends pilot signal  $\mathbf{s}_{gNB,m}$  in the coherence time;
6:   1-st antenna receives  $\mathbf{y}_{gNB,m \rightarrow 1}$ ;
7:   gNB estimates channel  $\hat{h}_{gNB,m \rightarrow 1} = \mathbf{y}_{gNB,m \rightarrow 1} (\mathbf{s}_{gNB,m})^\dagger$ ;
8:    $c_{gNB,m \rightarrow 1} = \frac{\hat{h}_{gNB,m \rightarrow 1}}{\hat{h}_{gNB,1 \rightarrow m}}$ ;
9: end for
10:  $\mathbf{C}'_{gNB} = \text{diag}(1, c_{gNB,2 \rightarrow 1}, \dots, c_{gNB,M \rightarrow 1})$ ;
11: return  $\mathbf{C}'_{gNB}$ ;

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We first obtain $\mathbf{C}_{gNB}$. Since $\mathbf{T}_{gNB}$ and $\mathbf{R}_{gNB}$ are diagonal arrays, $\mathbf{C}_{gNB}$ can be written as

$$\mathbf{C}_{gNB} = \text{diag} \left(\frac{t_{gNB,1}}{r_{gNB,1}}, \frac{t_{gNB,2}}{r_{gNB,2}}, \dots, \frac{t_{gNB,M}}{r_{gNB,M}} \right), \quad (15)$$

where, $t_{gNB,m}$ represents the transmitting impairment of the m -th antenna of the gNB, and $r_{gNB,m}$ represents the receiving impairment of the m -th antenna of the gNB. For ease of representation, we denote $c_{gNB,m} = \frac{t_{gNB,m}}{r_{gNB,m}}$, ($m \in \{1, 2, \dots, M\}$), and rewrite (15) as

$$\mathbf{C}_{gNB} = \text{diag} (c_{gNB,1}, c_{gNB,2}, \dots, c_{gNB,M}), \quad (16)$$

The insight of [34], [35], and [36] suggests that while obtaining the absolute RF impairments like $t_{gNB,m}$ or $r_{gNB,m}$ may be unfeasible, it is possible to derive the RF impairments relations between different antennas. To obtain the gNB's R-RFF, we employ a cyclic antenna switching strategy for transmission [34], where all the antennas at the gNB's side send pilot signals to each other.

The process of obtaining R-RFF of gNB is shown in Algorithm 1. It is important to note that the procedure described in Algorithm 1 is envisioned as an initial procedure. The resulting relative impairment matrix, $\mathbf{C}'_{gNB}$, is then cached by the gNB and utilized as a known reference in the real-time post-equalization process for all subsequent UE identification attempts. Furthermore, since $\mathbf{C}_{gNB}$ characterizes the gNB's own hardware, a single, pre-obtained $\mathbf{C}_{gNB}$ matrix is utilized by the gNB as a constant reference to identify all users within its service area; it is not a user-specific matrix.

As detailed in the algorithm, the process of obtaining R-RFF of gNB begins by selecting a reference antenna. The choice of reference antenna is theoretically arbitrary, as changing it results in a new feature vector that is simply a deterministic transformation of the original, preserving the underlying impairment information. However, the requirement for our system is consistency. Therefore, for our implementation, we consistently select the first antenna as the reference.

For each other antenna m , the gNB then estimates the reciprocal channels, $\hat{h}_{gNB,1 \rightarrow m}$ (from reference to antenna m)

and $\hat{h}_{gNB,m \rightarrow 1}$ (from antenna m to reference), by simulating pilot transmissions $\mathbf{s}_{gNB,1}$ and $\mathbf{s}_{gNB,m}$ between the antenna pair in both directions within the channel's coherence time. The RF impairment relation coefficient, $c_{gNB,m \rightarrow 1}$, is then obtained by taking the ratio of these two channel estimates, as given by:

$$c_{gNB,m \rightarrow 1} = \frac{\hat{h}_{gNB,m \rightarrow 1}}{\hat{h}_{gNB,1 \rightarrow m}} \approx \frac{r_{gNB,1} \tilde{h}_{gNB,m \rightarrow 1} t_{gNB,m}}{r_{gNB,m} \tilde{h}_{gNB,1 \rightarrow m} t_{gNB,1}}, \quad (17)$$

where, $\tilde{h}_{gNB,m \rightarrow 1}$ and $\tilde{h}_{gNB,1 \rightarrow m}$, $m \in \{2, \dots, M\}$ are true channel between the 1-th and the m -th antenna of gNB. They satisfy $\tilde{h}_{gNB,m \rightarrow 1} = \tilde{h}_{gNB,1 \rightarrow m}$. Therefore, (17) can be denoted by

$$c_{gNB,m \rightarrow 1} \approx \frac{r_{gNB,1} t_{gNB,m}}{r_{gNB,m} t_{gNB,1}} = \frac{c_{gNB,m}}{c_{gNB,1}}, \quad (18)$$

The absolute impairment of the reference antenna is $c_{gNB,1} = \frac{t_{gNB,1}}{r_{gNB,1}}$ and the absolute impairment of the m -th antenna is $c_{gNB,m} = c_{gNB,1} c_{gNB,m \rightarrow 1}$. Since we can only get relative values $c_{gNB,m \rightarrow 1}$ instead of absolute values $c_{gNB,m}$, we construct the available relative matrix as

$$\mathbf{C}'_{gNB} = \text{diag}(1, c_{gNB,2 \rightarrow 1}, \dots, c_{gNB,M \rightarrow 1}). \quad (19)$$

Then $\mathbf{C}'_{gNB}$ can be expressed by $\mathbf{C}_{gNB}$ as

$$\begin{aligned} \mathbf{C}'_{gNB} &= \frac{1}{c_{gNB,1}} \text{diag}(c_{gNB,1}, c_{gNB,1} c_{gNB,2 \rightarrow 1}, \dots, \\ &\quad c_{gNB,1} c_{gNB,M \rightarrow 1}) \\ &= \frac{1}{c_{gNB,1}} \text{diag}(c_{gNB,1}, c_{gNB,2}, \dots, c_{gNB,M}) \\ &= \frac{1}{c_{gNB,1}} \mathbf{C}_{gNB}. \end{aligned} \quad (20)$$

Since $\frac{1}{c_{gNB,1}}$ is a constant, the lemma 1 still works. Equation (13) can be written as

$$\begin{aligned} F\hat{\mathbf{H}}_U &\approx \frac{1}{c_{gNB,1}} \mathbf{W} \mathbf{C}_{gNB} \hat{\mathbf{H}}_U \\ &\approx \frac{1}{c_{gNB,1}} \mathbf{T}_{UE} (\mathbf{R}_{UE})^{-1} \\ &= \frac{1}{c_{gNB,1}} \mathbf{C}_{UE}. \end{aligned} \quad (21)$$

$\frac{1}{c_{gNB,1}} \mathbf{C}_{UE}$ is the proposed channel independent R-RFF which is used as the input of RFF extractor.

V. RFF TRAINING AND IDENTIFICATION

The training and identification of RFF are core processes within an RFF system. This process should extract channel-independent and highly discriminative RFFs from the received signals, and enable accurate device identification in complex multipath environments. In our system, training and identification data is generated and collected through 5G multi-antenna simulation system, and the extracted RFF features are correctly labeled.

A. RFF Extraction Process

1) *Signal Transmission*: We follow the new framework of post-equalization scheme proposed in Section III-B. The gNB first finds the $\mathbf{C}_{gNB}$ by antenna cycling sending proposed in Section IV-B. Then gNB sends downlink reference signals and UE sends uplink reference signals during the coherence time. The reference signal is initially subject to RF impairments introduced by the transmitting device, then propagates through the multipath channel, and finally experiences RF impairments from the receiving device.

2) *Preprocessing*: The initial step entails performing Symbol Time Offset (STO) estimation with STO compensation. Then, we estimate the CFO from the reference signal using the frequency domain CFO estimation method [37].

Assume that the upcoming OFDM symbol, denoted as $Y_{l+1}[k]$, can be expressed as the product of the previous OFDM symbol $Y_l[k]$ and the complex exponential term $e^{j2\pi\psi}$, which accounts for the inter-symbol phase drift caused by CFO. We obtain $Y_{l+1}[k] = Y_l[k] e^{j2\pi\psi} = H_l[k] X_l[k] e^{j2\pi\psi}$.

After mitigating the impact of multipath delay through the utilization of cyclic prefix (CP), the estimation of CFO can be accomplished by utilizing the frequency domain differential technique introduced by Moose [37],

$$\psi = \frac{1}{2\pi} \arctan \frac{\sum_{k=0}^{N_{SC}-1} \text{Im}[Y_l[k] Y_{l+1}[k]^*]}{\sum_{k=0}^{N_{SC}-1} \text{Re}[Y_l[k] Y_{l+1}[k]^*]}, \quad (22)$$

where N_{SC} is the total number of subcarriers used for the estimation. The CFO estimated with this method can be utilized as an auxiliary feature for identification. We compensate for the CFO in the frequency domain signal to obtain the preprocessed signal.

3) *RFF Extraction*: gNB and UE estimate the channel using the reference signals. UE sends result of downlink channel estimation as data to gNB, and the gNB employs the post-equalization function F to extract channel-robust RFF.

B. Comprehensive Simulation

To enhance the robustness of the RFF-based classification, we introduce diverse training conditions to ensure the model generalizes well to various channel impairments and noise levels. This is achieved by incorporating multipath propagation effects and additive noise, allowing the model to learn stable RF fingerprint features under real-world conditions [7], [28].

1) *Noise Effects*: In this paper, the noise effects for comprehensive simulation includes both multipath and noise. As the channel effects have already been eliminated, it essentially leverages the noise introduced by the channel for improvement.

The multipath is simulated by the TDL channel model [38]. TDL channel is commonly used to simulate different multipath channel environments. The TDL model has various configurations, including TDL-A (indoor scenarios where multipath components have small delays), TDL-B (outdoor environment with a direct line of sight (LOS)), TDL-C (urban environment with significant delay spread and rich multipath), TDL-D (rural open environment with larger delay spreads but

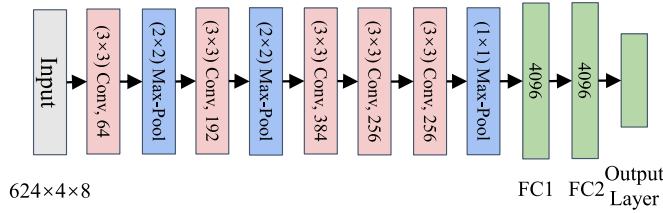


Fig. 3. Architecture of the classifier for R-RFF identification.

fewer multipath components), and TDL-E (extreme high-speed environments). In addition, the random seed of the same type of TDL channel is altered in each communication, which will change the fading coefficients, path delays, and gain powers for the multipath components, resulting in different realizations of the multipath channel.

We add different levels of noise to the training data to simulate the signal interference that may be encountered in the real world. In this way, the model can learn to extract effective RF fingerprint features in a noisy environment and improve the accuracy of recognition.

2) *Procedures*: The comprehensive simulation is carried out as follows:

- The multipath channel effect is generated with the different TDL channel model and random seed of the same type of TDL channel.
- During each signal transmission in Section V-A1, reference signal pass through the different channel.
- The artificial white Gaussian noise is added to the signal to emulate different SNR levels at the receiving side.

The comprehensive simulation process is applied to each training signals. The extracted channel-robust RFF have undergone comprehensive simulation. Since the RFF are extracted after post-equalization, the channel effects have already been removed, meaning that the comprehensive simulation introduces noise stemming from the channel. Unlike methods that aim to learn the invariant components of the RFF within the channel for identification or iterating over the channel, this approach is designed to combat the noise.

C. Model Architecture

The channel-independent R-RFF features can be regarded as a multi-channel 2-D image, which serves as the input to a convolutional neural network (CNN) for classification. Unlike end-to-end deep learning models that learn features directly from raw signals, our framework learns more abstract representations from the extracted R-RFF features to achieve effective class separability.

The model architecture is an adapted version of AlexNet [39], specifically modified to classify pre-extracted R-RFF features, which are structured as multi-channel 2D inputs. The architecture consists of two main components: a convolutional backbone that processes the structured R-RFF features and a fully connected classification head.

The CNN architecture is shown in Fig. 3. The convolutional backbone consists of five convolutional layers, one adaptive average pooling layer, and three fully connected layers. The

first convolutional layer employs 64 filters with a 3×3 kernel, followed by rectified linear unit (ReLU) activation and a max-pooling layer with a stride of 2. The second layer also uses 192 filters with a 3×3 kernel, followed by ReLU and another max-pooling layer. The third to fifth convolutional layers use 384, 256, and 256 filters, respectively, all with 3×3 kernels and ReLU activations. The final pooling layer is an adaptive average pooling layer that reduces the output to a 1×1 feature map.

The fully connected classification head consists of two layers with 4096 neurons each, both followed by ReLU and dropout for regularization, and a final output layer that matches the number of classification classes. This architecture is designed to classify the structured R-RFF features effectively, rather than extracting new features.

D. 5G RFF Wireless Protocol

We design a wireless protocol tailored to 5G systems to primarily include identity authentication for devices during the access phase, as well as seamless integration of periodic identification within regular communication.

1) *SRS and CSI-RS Description*: In the realm of wireless communication protocols, the successful implementation of UE identification and the extraction of RFF relies heavily on the intricate interplay between various reference signals. Sounding reference signal (SRS) and channel state information-reference signal (CSI-RS) are specially designed for uplink and downlink channel sounding in 5G [38]. They inherently contain comprehensive wireless channel information of both the device and environment, providing richer radio frequency feature information compared to demodulation reference signal (DMRS) and primarily/secondary synchronization signal (PSS/SSS). This abundant information endows CSI-RS and SRS with advantages in RFF. Additionally, they demonstrate great flexibility in that sounding can be executed only when needed by the gNB for channel estimation. When applied to identity recognition, this flexibility allows identity checks of UE to be conducted on an as-needed basis according to the gNB's requirements.

2) *Devices Initial Identification*: In 5G systems, all devices have to perform a random access process to access the gNB's resources [38]. In our protocol, identification is subsequent to the successful random access contention in 5G. We opt to undertake authentication subsequent to the successful random access contention and conduct CSI-RS and SRS transmission during the subsequent time slot in order to accomplish RFF authentication. Additionally, since the random access procedure only schedules a single antenna, the scheduling of multiple antennas can only be executed post-random access. Hence, we can schedule multiple antennas during this period to implement our proposed R-RFF authentication process. We make the most of the RF impairments in the transmission of the reference signals to accomplish the RFF.

The protocol is activated when a UE attempts to random access the BS in the network. 1) The BS continues to broadcast the broadcast signal. 2) The UE selects a random access preamble from a predefined set and sends it to the BS. 3) If the preamble is successfully received, the BS responds with a

random access response (RAR). 4) UE sends physical uplink shared channel (PUSCH) to request the establishment of a connection and request for the service. 5) BS sends contention resolution signal to UE. 6) BS employs multi-antennas and sends CSI-RS to the UE for downlink channel estimation. 7) Subsequently, within the coherent time, the UE responds by transmitting SRS back to the BS for uplink channel detection. 8) The UE then estimates the downlink channel based on the received CSI-RS, calculates the CSI and reports this information back to the BS.

3) *Devices Periodic Identification:* During the data communication between the gNB and the UE via physical downlink shared channel (PDSCH) and PUSCH, the channel is measured. Each periodic measurement triggers an RFF to continuously identify the device. If the identity of the device is verified as legitimate, data transmission will proceed seamlessly. Conversely, if the device's identity is found to be illegitimate, the connection will be promptly terminated to ensure security and integrity. The specific steps are outlined as follows: 1) The BS and UE transmit data to each other. 2) BS sends CSI-RS to the UE for downlink channel estimation. 3) Subsequently, within the coherent time, the UE responds by transmitting SRS back to the BS for uplink channel detection. 4) The UE then estimates the downlink channel based on the received CSI-RS, calculates the CSI and reports this information back to the BS. 5) If the identity of the device is verified as legitimate, data transmission will proceed seamlessly.

Furthermore, to avoid channel ageing, UE utilizes CSI-RS for channel estimation and compares it with the channel estimation results saved from the last authentication. UE calculates the coherence coefficient between the two channel estimation results. If the coherence coefficient $\rho \geq C$, the process directly proceeds to identify UE's identity, eliminating the need to transmit CSI. If $\rho < C$, the UE is required to transmit CSI. The UE calculates the CSI and reports this information back to the BS. Subsequently, BS employs the R-RFF algorithm to identify UE's identity.

E. Computational Complexity Analysis

To evaluate the feasibility of deploying our R-RFF framework in practical 5G systems, we analyze the computational complexity of each key component.

The process begins with the calculation of the post-equalization matrix $\mathbf{W} = (\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T)^{-1} \hat{\mathbf{H}}_D^*$, where the dominant cost is the inversion of the $N \times N$ matrix $\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T$, resulting in a complexity of $\mathcal{O}(N^3)$. Given that N is small in our system (typically 2 or 4), this step is computationally efficient. Subsequently, the R-RFF feature is extracted via the matrix multiplication $\mathbf{W} \mathbf{C}_{\text{gNB}} \hat{\mathbf{H}}_U$, which has an overall complexity of $\mathcal{O}(N^2 M)$. With $M = 32$ and small N , this matrix operation can be executed efficiently on baseband processing units. Finally, the resulting R-RFF features input tensor is fed into our lightweight AlexNet-style CNN for classification, with 5 convolutional layers and 2 fully connected layers. The total number of multiply-accumulate (MACs) operations for a single inference is estimated to be less than 10^6 . In comparison,

end-to-end models that process raw IQ data often require computational loads exceeding 10^8 MACs [6].

F. Security Analysis

This section analyzes the resilience of the proposed R-RFF scheme against sophisticated adversarial attacks, such as impersonation, model-based forgery, and replay attacks. Our defense is multi-layered, relying on the physical, protocol, and operational characteristics of the system.

1) *Threat Model:* We consider a probabilistic polynomial-time (PPT) adversary. The adversary's goal is to impersonate a legitimate UE by transmitting a forged signal, thereby gaining unauthorized access. The adversary can eavesdrop on all wireless communications between UE and gNB, inject arbitrary signals to channels, and has full knowledge of the R-RFF protocol and algorithms, but cannot know the precise internal hardware impairment values of legitimate devices.

2) *Physical Layer Resilience:* At a fundamental level, the security of our scheme is rooted in the R-RFF being a form of physical unclonable function (PUF). The RFF $\mathbf{C}_{UE}$ is derived from minute, random manufacturing variations in the UE's RF components. To forge this RFF, an adversary would need to perfectly characterize and replicate the internal hardware impairments of both the target UE ($\mathbf{T}_{UE}, \mathbf{R}_{UE}$) and the gNB $\mathbf{T}_{\text{gNB}}$, which is considered practically infeasible.

3) *Protocol-Level Security:* As a foundational defense, the CSI feedback loop is protected by TS 33.501 [40]. The uplink transmission of the $\hat{\mathbf{H}}_D$ matrix on the PUSCH is presumed to occur after the access stratum security context has been established. This ensures the feedback data is protected by 5G's native encryption and integrity protection mechanisms, mitigating trivial eavesdropping and manipulation attacks by external parties. In addition, we do not consider pilot contamination as a problem, and we assume it to be mitigated by existing approaches in the literature [41].

4) *Resilience Against Forgery and Replay Attacks:* Even if an adversary could bypass protocol-level encryption, forging a valid R-RFF in real-time remains infeasible due to several operational barriers:

- **Replay Attack Barrier:** In a dynamic environment, the propagation channel $\hat{\mathbf{H}}_D$ changes over time and position. A previously intercepted $\hat{\mathbf{H}}_D$ feedback would be outdated and invalid for the gNB's current post-equalization process, rendering simple replay attacks ineffective.
- **Eavesdropping Barrier:** An adversary faces a fundamental obstacle when trying to obtain clean information: their own hardware impairments. Any signal intercepted by the adversary is inevitably convoluted with their own receiver's unique RF impairments $\mathbf{R}_{\text{Ev}}$. This "blinding" effect prevents them from acquiring the clean, uncontaminated data needed to accurately analyze or reverse-engineer the target's fingerprint.
- **Forgery Barrier:** Assuming an adversary could overcome the previous barriers and obtain a clean copy of the CSI feedback, $\hat{\mathbf{H}}_D^{(\text{target})} = \mathbf{R}_{UE}^{(\text{target})} \hat{\mathbf{H}}_D^{(\text{target})} \mathbf{T}_{\text{gNB}}$, forging the true RFF is still infeasible. The adversary needs to construct $\mathbf{C}_{UE}^{(\text{target})} = \mathbf{T}_{UE}^{(\text{target})} (\mathbf{R}_{UE}^{(\text{target})})^{-1}$, which requires knowledge of both the target's transmit $\mathbf{T}_{UE}^{(\text{target})}$ and receive $\mathbf{R}_{UE}^{(\text{target})}$.

TABLE III
SIMULATION SETUP

Parameters	Value
The number of wireless frames	528,000
The number of UEs	110
The number of UE antennas N	2 / 4
The number of gNB antennas M	32
Subcarrier frequency spacing Δf	15 kHz
Carrier frequency f_c	3.072 GHz
Bandwidth B	15 MHz
Channel model	AWGN / Rician / TDL

impairments. The eavesdropped $\hat{\mathbf{H}}_D^{(target)}$ contains no information about $\mathbf{T}_{UE}^{(target)}$ and does not allow for the unique isolation of $\mathbf{R}_{UE}^{(target)}$ from the other unknown variables. Lacking these essential components, the adversary cannot construct the target's fingerprints.

In summary, our R-RFF scheme serves as a powerful complementary security layer. While cryptographic methods secure a device's digital identity, R-RFF provides a strong binding to its physical hardware. This offers enhanced protection against sophisticated threats such as device cloning and identity theft, and provides an orthogonal security dimension that strengthens the overall trust model, especially against future threats like quantum computing-enabled adversaries that may break currently deployed cryptographic solutions. Our approach is lightweight on the UE hardware but shifts complexity to the gNB.

VI. SIMULATION EVALUATION

In this section, we evaluate the performance of our proposed R-RFF in terms of classification and multipath channels resistance via Matlab simulations.

A. Simulation Setup

This section describes the simulation settings used to evaluate the proposed method. We first introduce the general system parameters, including the network architecture and resource configurations. Then, we detail the reference signal settings, RF impairments, and channel models adopted in our simulations. Finally, we outline the dataset preparation, including training, validation, and testing splits.

We use MATLAB 5G NR Toolbox to simulate 110 UEs. As shown in Table III, the gNB is assumed to have $M = 32$ antennas, and the different UEs are assumed to have $N = 2$ or 4 antennas. Both the antennas of gNB and UEs are assumed to be fixed antennas similar to [34]. The length of a complete 5G NR wireless frame is fixed at 10 ms. The length of the subframe is also fixed at 1 ms. Each wireless frame contains 10 subframes. The carrier frequency f_c is set to 3.072 GHz, subcarrier spacing (SCS) Δf is set to 15 kHz, the bandwidth B is set to 15 MHz, and 948 subcarriers are implemented. The uplink employs SRS and the downlink employs CSI-RS as the pilot signal.

1) *Reference Signal Configuration*: According to 3GPP 38.211 [38], the SRS can be located in the last 6 symbols in a slot with consecutive {1, 2, 4} symbols. We set 4 consecutive

TABLE IV
RF IMPAIRMENTS CONFIGURATION

Parameters	Settings
IQ imbalances	All the UEs and gNB have IQ imbalances. The ranges of gain and phase imbalances are set as [-1, 1] dB and [-5, 5] degrees respectively.
CFO	All the UEs and gNB have CFO and the CFO is the same on all antennas from the same UE. The ranges of CFO are set as [-15, 15] kHz.
Phase Noise	All the antenna chains have phase noise and the ranges of phase noise are set as [-80, -100] dBc/Hz.
PA nonlinearity	All the UEs and gNB have PA nonlinearities. Each parameter in polynomial coefficients of the PA nonlinearity was varied within $\pm 5\%$ of the measured values.

symbols in the time domain in this paper. They are allocated on the resource grid under a particular antenna. We assume that the SCS is 15 kHz, so there will be 14 OFDM symbols in a time slot. Assuming that the length of each OFDM symbol is T_{sym} , the total time of the SRS duration portion of an uplink subframe containing SRS is $4T_{sym}$. Since the 4 SRS sequences are generated using the same method, they are identical and have a consistent interval of T_{sym} between them, which can be used to extract CFO. The SRS's frequency position at the start of the bandwidth part (BWP) is set to 0, and no additional frequency offset is employed, ensuring the SRS's alignment with the beginning of the BWP in resource blocks (RBs).

In terms of CSI-RS, the CSI-RS is configured with the following settings: We set the type of CSI-RS as non-zero power (NZP), set the transmission to occur every fifth frame with an offset of one frame for regular updates, and choose a density of 'one' to ensure comprehensive coverage. The CSI-RS is precisely transmitted on the 2nd and 6th OFDM symbols within each slot.

2) *RF Impairments Configuration*: A comprehensive set of RF hardware impairments was modeled for each transmit and receive chain of the gNB and UEs. The specific parameterization was chosen to be representative of typical 5G systems, guided by 3GPP standards. The configuration is shown in Table IV. We configure each RF chain of the gNB and UEs with different IQ imbalances. The hardware chains of all antennas for the same device have the same CFO, because different commercial RF front-ends often share a common oscillator. We set the ranges of gain and phase imbalances as [-1, 1] dB and [-5, 5] degrees, respectively [42], [43]. For each antenna of every UE and gNB, a fixed value is randomly selected from the given range to represent the IQ imbalances associated with that antenna. We set the ranges of CFO as [-15, 15] kHz [44]. For different UE, we set a fixed value randomly selected from the given range. For all antennas of the same UE, we set the same value to represent the CFO associated with that UE. The phase noise level is modeled to vary randomly within the range of [-80, -100] dBc/Hz. We use the NXP Airfast LDMOS Doherty PA provided in the MATLAB Communications Toolbox named Power Amplifier Characterization [45]. The generalized memory polynomial (GMP) model of PA was used to import PA nonlinearities. We measured PA input and output data to model the PA.

TABLE V
CHANNEL CONFIGURATION

Channels	Settings
AWGN	There's only one direct path.
Rician	There is a LOS path and a reflected path, whose delays are set to $[0 \ 30e-9]$ s, average path gains are set to $[-3 \ -5]$ dB, and Rician factor is set to 3 dB.
TDL-C	The TDL-C channel consists of a total of 24 NLOS paths. The delays range from 0.2099 to 8.6523 ns, and the path gains vary from -22.8 dB to 0 dB. The fading distribution for all paths follows a Rayleigh distribution.

The memory length and the polynomial degree are both set to 5. Each parameter in polynomial coefficients of the PA nonlinearity was varied within $\pm 5\%$ of the measured values. Furthermore, to provide a holistic validation of the modeled impairments, the aggregate effect of these non-idealities was controlled such that the overall error vector magnitude (EVM) of the simulated transmitted signals complies with the limits specified for 5G UEs in 3GPP TS 38.104 [46].

3) *Channel Configuration*: As shown in Table V, AWGN channel, Rician channel, and TDL channel in 3GPP TR 38.901 [32] are adopted for simulation. AWGN channel has only one direct path. The Rician channel has 2 paths, namely the LOS path and the non-line of sight (NLOS) path, whose delays are set to $[0 \ 30e-9]$ s, the average path gains are set to $[-3, -5]$ dB, and the Rician factor is set to 3 dB. The TDL-C channel, a standard model for challenging urban NLOS environments, is used extensively in our evaluation. It features 24 paths with distinct delays and power levels. The delays range from 0.2099 to 8.6523 ns, and the path gains vary from -22.8 to 0 dB, as specified in the 3GPP protocol, representing a rich multipath scenario. The fading distribution for all paths follows a Rayleigh distribution. In the simulation, the delay spread and Doppler shift are configured based on the deployment scenario. For instance, TDL-A is configured with a delay spread of $10e-9$ s and a maximum Doppler shift f_m of 10 Hz, representing a low-mobility indoor environment. Other TDL models are configured according to the recommended settings in 3GPP TR 38.901, ensuring realistic multipath characteristics across different propagation conditions. For each SNR level, the multipath parameters, including path delays and gains, are regenerated using a newly initialized random seed, ensuring statistical variability across different realizations.

4) *Training and Classification*: The dimensions of the extracted channel-independent R-RFF feature set are configurable and depend on the system parameters. The overall feature structure is an $N_{SC} \times N \times N$ tensor, where N_{SC} is the number of subcarriers used for extraction and N is the number of UE antennas. According to our simulation settings, the channel-independent R-RFF has a size of $624 \times 4 \times 4$, where 4×4 represents the extracted R-RFF features per channel, and 624 denotes the total number of channels. We employ the CNN for the training of the extracted R-RFF. The input to the network is constructed by first stitching the real and imaginary parts of the $N_{SC} \times N \times N_{UE}$ into an matrix of $N_{SC} \times N \times 2N$. We put the preprocessed array into the CNN model for training. To ensure reliable model

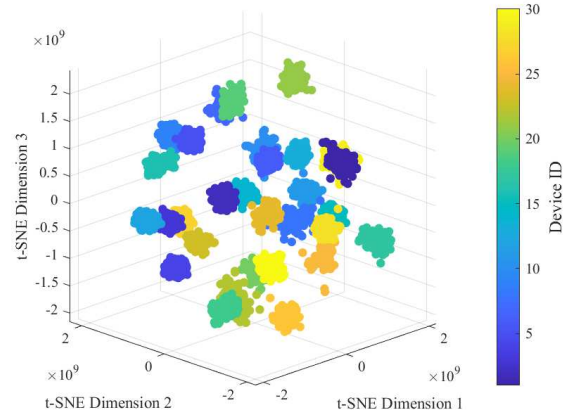


Fig. 4. R-RFF features t-SNE feature visualization.

evaluation and prevent overfitting, we split the dataset into training, validation, and testing sets in a 6 : 2 : 2 ratio. Specifically, for each device, we collected 316,800 frames for training, 105,600 frames for validation, and 105,600 frames for testing. The data were generated at discrete SNR points from -5 dB to 30 dB to evaluate performance across a wide range of channel quality conditions.

The classification stage follows a similar flow to the training stage, with one key difference: the gNB no longer generates its own R-RFF features. Instead, these features were already generated and stored during training. During wireless communication, the UE and gNB execute a wireless identification protocol and exchange pilot signals. The UE sends back the channel estimation results, and the gNB extracts both R-RFF and CFO for identification. To determine the UE's identity, the gNB inputs the extracted features into the trained CNN model for classification. The predicted label is then compared with the ground-truth label of the device under test to evaluate the performance and robustness of the proposed R-RFF scheme.

B. Simulation Results

1) *Stability of R-RFF*: To evaluate the stability and discriminability of the features extracted by the R-RFF method, we randomly generated thirty devices and employed the t-distributed stochastic neighbor embedding (t-SNE) technique to project the entire feature space into three dimensions for visualization. Fig. 4 presents the t-SNE result for features extracted from 30 distinct devices under the TDL-C channel at 30 dB SNR. In the plot, each color corresponds to the features from a single device. It is clearly observable that the features from different devices form dense and well-separated clusters, with minimal overlap between them.

2) *Effects of the Number of UEs*: To evaluate the impact of the number of devices on the robust R-RFF scheme, we simulated the classification accuracy for different numbers of devices, ranging from 10 to 100, under the TDL-C channel. The results of Fig. 5 demonstrate that the classification accuracy of the robust R-RFF scheme improves as the SNR increases but decreases as the number of devices increases.

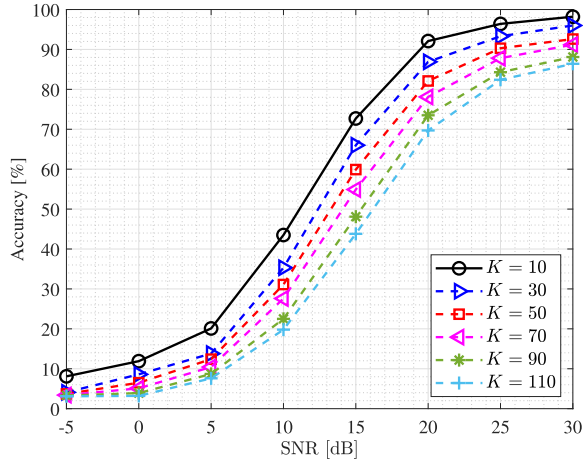


Fig. 5. Classification accuracy of the different number of UEs ($N = 4$, $M = 32$).

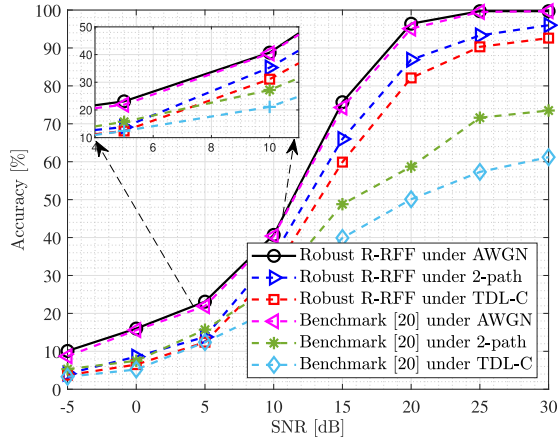


Fig. 6. Classification results on different methods ($N = 4$, $M = 32$, $K = 30$).

Specifically, for a small number of devices (10 devices), the accuracy surpasses 98.2% when the SNR is 30 dB, while achieving 72.7% even at a moderate SNR of 15 dB. However, as the number of devices increases, the classification accuracy decreases. For instance, when the number of devices reaches 50, the accuracy drops to 59.9% at 15 dB and 90.3% at 25 dB.

When the number of devices is 90 or more, the accuracy exhibits a significant decline, with the performance at SNR = 10 dB decreasing to 22.5% for 90 devices and 19.8% for 110 devices. At high SNR values, the classification accuracy remains relatively high, with 110 devices achieving 86.4% at 30 dB. These results indicate that while the robust R-RFF scheme performs well in high-SNR scenarios, the classification accuracy decreases with an increasing number of devices, highlighting the challenge of distinguishing a larger number of devices in practical applications.

3) *Robustness and Multipath Resistance of R-RFF Scheme:* To evaluate the robustness and multipath resistance of the proposed robust R-RFF scheme, we compare its classification accuracy with the benchmark [20] under different channel conditions, including AWGN, a two-path channel, and the TDL-C channel. The classification accuracy across these scenarios is presented in Fig. 6, where we focus on the performance for 30 devices as an example.

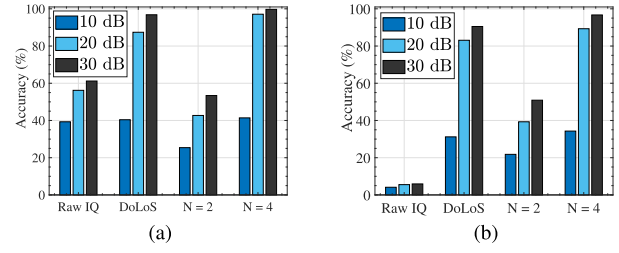


Fig. 7. Comparison between single-antenna method and R-RFF. (a) Results under AWGN channel. (b) Results under TDL-C channel.

As shown in the results, both methods perform well in the AWGN channel. However, in the presence of multipath fading, the benchmark method exhibits a considerable performance degradation. Specifically, its accuracy drops to 71.6% in the two-path channel and further declines to 57.3% in the TDL-C channel at 25 dB, indicating its vulnerability to complex channel effects. The results are likely attributable to two reasons. Firstly, method [20] is designed to exploit relative phase differences from independent RF oscillators. This type of inter-oscillator phase offset is not a prominent feature in our common-oscillator simulation model. Secondly, the benchmark's reliance on the assumption that all antennas experience an identical multipath channel, which allows for cancellation via subtraction, is challenged by our use of a realistic TDL-C NLOS model. The spatial variations in the TDL-C model likely lead to residual multipath interference, which can contaminate the features extracted by the benchmark method.

In contrast, the proposed R-RFF scheme demonstrates superior multipath resistance. Under the same conditions, its classification accuracy for 30 devices remains at 93.3% in the TDL-C channel and 86.9% in the two-path channel at 25 dB. Even at a lower SNR of 15 dB, the R-RFF method still achieves 59.9% accuracy in the TDL-C channel, whereas the baseline method only reaches 39.9%. These results confirm the robustness of the proposed R-RFF scheme in handling multipath environments and its significant improvement over traditional baseline approaches.

To further compare the performance of single-antenna and our R-RFF scheme, we evaluate raw IQ [6] and DoLoS [25]. The results, evaluated under both AWGN channel and TDL-C channel at SNRs of 10, 20, and 30 dB with 20 UEs, are shown in Fig. 7.

Both the DoLoS method and our R-RFF scheme significantly outperform the raw IQ baseline, confirming that robust feature extraction is essential in multipath environments. Notably, our 2-antenna R-RFF configuration, with its limited single relative feature, shows lower accuracy than the specialized DoLoS method. However, by increasing to a 4-antenna configuration, our scheme's performance becomes comparable to the single-antenna approach. This highlights the core principle of our work: leveraging multiple antennas unlocks a new degree of freedom for fingerprinting through inter-antenna relative impairments.

4) *Effects of the Number of Antennas:* To evaluate the impact of different antenna configurations on the identification performance of the R-RFF method, we conducted simulations with two-antenna and four-antenna setups. The classification

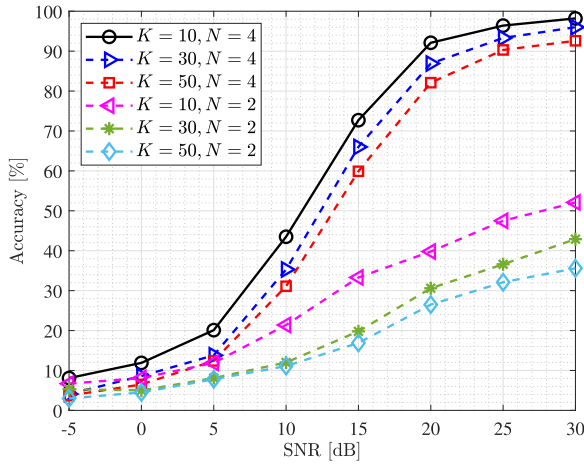


Fig. 8. Classification accuracy of the different number of antennas ($M = 32$, R-RFF method).

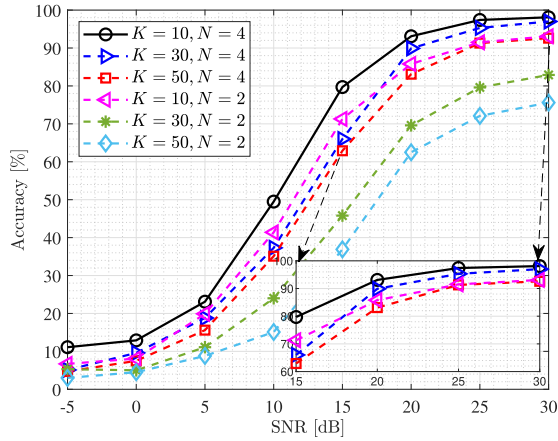


Fig. 9. Classification accuracy of the different number of antennas ($M = 32$, CFO-assisted robust R-RFF method).

accuracy results for different numbers of antennas are shown in Fig. 8. The classification accuracy with four antennas is consistently higher than that with two antennas across all SNR levels. For example, when identifying 10 devices, the four-antenna R-RFF scheme achieves 72.7% accuracy at 15 dB, whereas the two-antenna scheme only reaches 33.3%. Similarly, at 30 dB, the accuracy for four antennas reaches 98.2%, whereas it remains at 52.1% for two antennas. This performance gap is also observed for larger numbers of devices. When classifying 50 devices at 20 dB, the accuracy with four antennas is 82.1%, while it drops to 26.5% with two antennas.

This performance gap arises because the proposed R-RFF scheme leverages the impairment relation between antenna chains for classification. In a two-antenna device, only one antenna can be used as a feature, limiting the available impairment information and resulting in lower classification accuracy. In contrast, with four antennas, more impairment relations are available, leading to improved classification performance. These results suggest that this scheme which relies on the inter-antenna relation will become more and more effective in the future as the number of antennas of the UEs increases.

Considering the poor classification of the R-RFF method when the number of antennas is small, we propose the

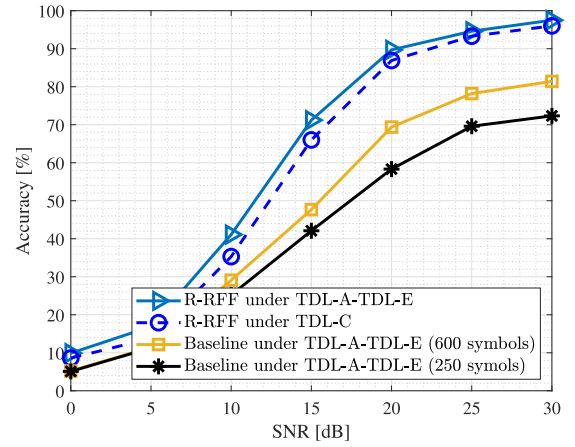


Fig. 10. Classification accuracy of the R-RFF features with and without comprehensive simulation ($N = 4$, $M = 32$, $K = 30$).

CFO-assisted robust R-RFF method. The CFO extraction can be achieved with only one antenna. Hence, when the number of antennas is limited, the CFO can have a more significant impact on UE classification. Meanwhile, different antennas of the same device share a crystal oscillator, so the CFOs on different antennas are the same. Therefore, as the number of antennas increases, the potential benefits provided by R-RFF surpass those of the CFO. The combination of these two approaches not only retains the advantages of R-RFF in multi-antenna devices but also mitigates the dependency of the R-RFF method on the number of antennas. In order to compare the effect of different antenna numbers on the identification achieved by the CFO-assisted robust R-RFF method and R-RFF method, we simulated the UEs classification accuracy with different antenna numbers under two methods. We used two-antenna and four-antenna for simulation analysis. Fig. 9 presents the classification accuracy with the CFO-assisted robust R-RFF method under different antenna numbers. The robust RFF method consistently upholds a classification accuracy rate exceeding 80% across both dual and quadruple antenna configurations, when subjected to SNR levels exceeding 25 dB. This improves the low accuracy with 2 antennas observed in the results presented in Fig. 8. Furthermore, Fig. 9 illustrates a disparity in performance between the four-antenna and two-antenna scenarios, thereby affirming our initial hypothesis that an increase in the number of antennas corresponds to an improvement in classification performance.

5) *Comparison of R-RFF and IQ-Based Classification Under Different Training Conditions:* The purpose of Fig. 10 is to analyse how different training conditions affect the accuracy of R-RFF, as well as the accuracy of the RFF method based on IQ signal classification, which uses the same training conditions but without extracting anti-multipath features. The simulation setup includes variations in noise levels, channel realizations, and TDL model types to ensure a diverse evaluation environment. The evaluation is conducted for a scenario with 30 UEs and four antennas.

As the SNR increases from 0 to 30 dB, R-RFF trained under diverse channel conditions consistently demonstrates higher accuracy compared to models trained with limited

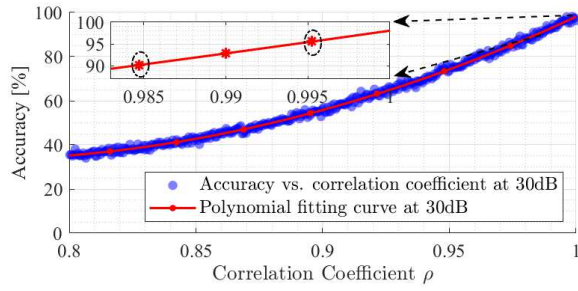


Fig. 11. Fitting analysis of correlation coefficient and identification accuracy.

channel and noise variations. The multipath channel effects are eliminated when extracting the R-RFF features. Therefore, changes in the channel through comprehensive simulation do not cause interference in the identification process. Instead, these changes introduce noise that improves the accuracy of R-RFF features at low SNR.

Additionally, IQ-based classification baselines [7] are evaluated with 256 and 600 symbols. While using 600 symbols improves accuracy to 81.41% at 30 dB compared to 72.33% with 256 symbols, both IQ-based methods still perform worse than R-RFF. At 10 dB SNR, the best IQ-based method achieves only 47.67% accuracy, whereas R-RFF trained with diverse channel conditions reaches 71.2%. This highlights that while increasing the number of symbols can improve IQ-based classification, it remains fundamentally limited by the lack of robust feature extraction.

6) *Impact of Doppler Effects on Identification Accuracy:* The effectiveness of our proposed scheme hinges on the channel reciprocity assumption ($\hat{\mathbf{H}}_U \approx \hat{\mathbf{H}}_D^T$) in TDD systems. In mobile scenarios, however, this assumption is challenged by channel time-variation due to Doppler effects, which can degrade identification accuracy. To quantify this impact, we first analyze the functional relationship between accuracy and the channel's temporal correlation coefficient ρ . The two downlink channel estimation results at the t_a moment and the t_b moment can be represented as $\hat{\mathbf{H}}_D(t_a)$, $\hat{\mathbf{H}}_D(t_b)$ and $t_b = t_a + \tau$. We reshape two $M \times N$ dimensional matrices into $1 \times MN$ sequences and calculate the correlation coefficient between these two sequences. We adopt the Jakes Doppler spectrum model for this analysis as it provides a precise mathematical relationship between ρ , the maximum Doppler shift f_m , and the inter-measurement time delay τ , given by

$$\rho(\hat{\mathbf{H}}_D(t_a), \hat{\mathbf{H}}_D(t_b)) = \frac{E\{\hat{\mathbf{H}}_D(t_a)^H \hat{\mathbf{H}}_D(t_a + \tau)\}}{\sigma_h^2} = J_0(2\pi f_m \tau), \quad (23)$$

where, σ_h^2 is the variance of channel gain and $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind.

To establish this relationship, we conducted over 500 simulation runs at an SNR of 30 dB. We systematically varied the correlation coefficient ρ by adjusting the product of the Doppler shift f_m and time delay τ . The results, plotted in Fig. 11, reveal a strong positive dependence of the identification accuracy on the channel correlation. The inset highlights this critical relationship: to achieve an accuracy of 90%, a correlation coefficient of $\rho \approx 0.982$ is required. Achieving a

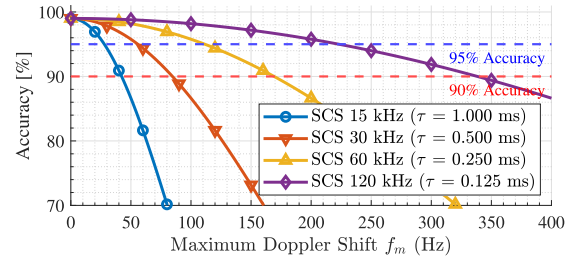


Fig. 12. Accuracy vs. Doppler shift for different SCS configurations.

higher accuracy of 95% demands an even stricter correlation of $\rho \approx 0.992$.

Having established these correlation requirements, we translate them into practical operational limits in Fig. 12. This figure illustrates the identification accuracy as a function of Doppler shift for various 5G NR SCS configurations, where the time delay τ corresponds to one slot length. To maintain 95% accuracy, 15 kHz SCS configuration ($\tau = 1.0$ ms) can only tolerate a maximum Doppler shift of 28 Hz, corresponding to a velocity of 9.7 km/h at a 3.072 GHz carrier frequency, accommodating pedestrian scenarios. By contrast, a 120 kHz SCS ($\tau = 0.125$ ms) can support speeds up to 78 km/h (223 Hz) for the same accuracy level.

If the accuracy requirement is relaxed to 90%, the system's robustness increases. The 15 kHz SCS can support speeds up to 14.9 km/h (42 Hz), while the 120 kHz SCS can accommodate highway speeds of approximately 119 km/h (338 Hz). This analysis provides a quantitative framework for deploying our scheme in mobile environments. It validates that by appropriately selecting the SCS, operators can effectively manage the trade-off between security performance and supported user mobility, ensuring the scheme's viability across diverse scenarios, from static IoT devices to vehicular UEs.

VII. LIMITATIONS AND FUTURE WORK

While our proposed R-RFF scheme demonstrates significant potential, we acknowledge several practical limitations and corresponding directions for future research.

A. Practical Deployment and Scalability

1) *Scalability and Uniqueness in Practical Deployments:* We envision our R-RFF scheme as a security enhancement for scenarios like private 5G networks (e.g., for industrial or enterprise use). However, a broader challenge for any RFF application is to demonstrate statistical distinguishability across large-scale deployments of near-identical chipsets.

2) *Training Data Requirements and Efficiency:* The total amount of training data used in our comprehensive evaluation is substantial. This aggregate dataset was created across 48 distinct channel and SNR conditions to ensure the resulting model is highly robust. For a practical deployment, the required data per device is significantly smaller (approx. 6,600 frames, taking about one minute to collect), making the enrollment process feasible. Future work will further explore data-efficient methods like few-shot learning to minimize this requirement.

B. Robustness Under Diverse and Adverse Conditions

1) *Experimental Validation and Long-Term Stability:* While this paper's evaluations are based on comprehensive

simulations designed to reflect realistic 5G conditions, experimental validation on a real-world hardware testbed is an essential next step. This future work should serve two primary goals: first, to verify the scheme's immediate practical performance and robustness, and second, to conduct a long-term analysis of feature stability. The stability analysis is crucial for characterizing the effects of hardware aging and developing compensation mechanisms to ensure reliability over a device's lifetime.

2) *Robustness in Low-SNR and High-Interference Conditions*: Furthermore, while this work establishes a framework for channel robustness against multipath, maintaining high identification accuracy in low-SNR and high-interference environments remains a significant challenge. These adverse conditions can corrupt the underlying channel estimates and obscure the subtle hardware features. Therefore, a critical direction for future work is the development of novel signal processing and machine learning techniques specifically designed to enhance RFF resilience in these scenarios.

C. Security and Privacy Analysis

1) *Security Considerations*: From a security perspective, a key limitation of this work is the lack of a formal, information-theoretic security analysis, which remains a vital direction for future research. Although the current literature on RFF predominantly focuses on empirical robustness, a rigorous assessment would require the derivation of theoretical metrics, such as the mutual information between legitimate and optimally forged fingerprints, to formally quantify the security guarantees of physical layer authentication systems.

2) *Privacy Considerations*: From a privacy perspective, a potential concern is the leakage of sensitive information, such as user grouping, through correlated fingerprints. While our channel decoupling framework is designed to mitigate this risk by making the fingerprint location-independent, future work could further enhance privacy. Techniques like active fingerprinting, could be developed to formally prevent side-channel clustering attacks.

VIII. CONCLUSION

In this paper, we proposed a multi-antenna channel-robust RFF method to address the two bottleneck problems of RFF in the multi-antenna system, i.e., multipath channels and imperfect channel reciprocity. While precoding can remove ideal multipath channels via channel echoing, imperfect channel reciprocity makes this approach infeasible. To overcome this deficiency, we proposed a post-equalization framework to utilize CSI feedback to counteract the multipath channels. By establishing the RF impairment relation between the antenna chains of a multi-antenna device, we extracted a stable channel-robust RFF that eliminates the non-reciprocal portion of the uplink and downlink channels. The simulation results showed that the proposed robust R-RFF scheme can identify RFF effectively in multipath propagation scenarios in a TDD multi-antenna system.

APPENDIX A

Assume that $\hat{\mathbf{H}}_U = \mathbf{R}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}$ and $\hat{\mathbf{H}}_D = \mathbf{R}_{UE} \tilde{\mathbf{H}}_D \mathbf{T}_{gNB}$ respectively represents the channel estimation of uplink from

UE to gNB and downlink from gNB to UE, the designed post-equalization matrix $F = \mathbf{W} \mathbf{C}_{gNB}$ multiplies $\hat{\mathbf{H}}_U$ to obtain

$$F \hat{\mathbf{H}}_U \approx \mathbf{W} \mathbf{C}_{gNB} \hat{\mathbf{H}}_U, \quad (24)$$

where, the pseudo-inverse matrix of $\hat{\mathbf{H}}_D$ is denoted by

$$\mathbf{W} = (\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T)^{-1} \hat{\mathbf{H}}_D^*, \quad (25)$$

and the R-RFF of gNB is denoted by

$$\mathbf{C}_{gNB} = \mathbf{T}_{gNB} (\mathbf{R}_{gNB})^{-1}. \quad (26)$$

Substituting (25) into (24) and further expanding $\hat{\mathbf{H}}_D$ and $\hat{\mathbf{H}}_U$ in (24) yields

$$\begin{aligned} F \hat{\mathbf{H}}_U &\approx (\hat{\mathbf{H}}_D^* \hat{\mathbf{H}}_D^T)^{-1} \hat{\mathbf{H}}_D^* \mathbf{C}_{gNB} \hat{\mathbf{H}}_U \\ &= (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{R}_{UE} \tilde{\mathbf{H}}_D \mathbf{T}_{gNB})^T)^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) \mathbf{C}_{gNB} (\mathbf{R}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}). \end{aligned} \quad (27)$$

Substituting (26) into (27) yields

$$\begin{aligned} F \hat{\mathbf{H}}_U &\approx (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{R}_{UE} \tilde{\mathbf{H}}_D \mathbf{T}_{gNB})^T)^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} (\mathbf{R}_{gNB})^{-1}) (\mathbf{R}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}). \end{aligned} \quad (28)$$

Since $(\mathbf{R}_{gNB})^{-1}$ is the inverse matrix of $(\mathbf{R}_{gNB})$, multiplying $(\mathbf{R}_{gNB})^{-1}$ and $(\mathbf{R}_{gNB})$ gives the unit array $\mathbf{I}$. The (28) can be rewritten as

$$\begin{aligned} F \hat{\mathbf{H}}_U &\approx (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{R}_{UE} \tilde{\mathbf{H}}_D \mathbf{T}_{gNB})^T)^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}) \\ &= (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{T}_{gNB}^T \tilde{\mathbf{H}}_D^T \mathbf{R}_{UE}^T))^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}). \end{aligned} \quad (29)$$

As a result that $\mathbf{T}_{gNB}$ is a diagonal array, $\mathbf{T}_{gNB}^T = \mathbf{T}_{gNB}$. Meanwhile, the uplink and downlink channels are reciprocal in the coherence time, which means $\tilde{\mathbf{H}}_D^T = \tilde{\mathbf{H}}_U$. In addition, $\mathbf{R}_{UE}$ is a diagonal array and invertible, which means $\mathbf{R}_{UE}^T = \mathbf{R}_{UE}$ and $\mathbf{R}_{UE} \mathbf{R}_{UE}^{-1} = \mathbf{R}_{UE}^{-1} \mathbf{R}_{UE} = \mathbf{I}$. Therefore (29) can be written as

$$\begin{aligned} F \hat{\mathbf{H}}_U &\approx (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{T}_{gNB}^T \tilde{\mathbf{H}}_D^T \mathbf{R}_{UE}^T))^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}) (\mathbf{R}_{UE}^T)^{-1} \mathbf{T}_{UE}. \end{aligned} \quad (30)$$

As

$$\begin{aligned} \mathbf{I} &= (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^* (\mathbf{T}_{gNB}^T \tilde{\mathbf{H}}_D^T \mathbf{R}_{UE}^T))^{-1} \\ &\quad (\mathbf{R}_{UE}^* \tilde{\mathbf{H}}_D^* \mathbf{T}_{gNB}^*) (\mathbf{T}_{gNB} \tilde{\mathbf{H}}_U \mathbf{T}_{UE}), \end{aligned} \quad (31)$$

The (30) can be expressed as

$$F \hat{\mathbf{H}}_U \approx (\mathbf{R}_{UE}^T)^{-1} \mathbf{T}_{UE}, \quad (32)$$

where we eliminate the influence of multipath channels and RF impairments of gNB, and obtain the features of UE. Since $\mathbf{R}_{UE}$ and $\mathbf{T}_{UE}$ are diagonal arrays, we can obtain

$$(\mathbf{R}_{UE}^T)^{-1} \mathbf{T}_{UE} = \mathbf{T}_{UE} \mathbf{R}_{UE}^{-1}. \quad (33)$$

Eventually, we obtain

$$F \hat{\mathbf{H}}_U \approx \mathbf{T}_{UE} \mathbf{R}_{UE}^{-1}. \quad (34)$$

This completes the proof of the lemma 1. $\blacksquare$

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